

ARCH 594/508: AI Research Seminar/Studio—AI in Architecture
Spring 2026

Generative AI

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TODAY'S AGENDA

1. Generative AI
2. Lab Activity: (Generative AI)

WHAT IS GENERATIVE AI?

- Generative AI refers to models that **learn patterns from data and produce new outputs that resemble that data.**
- Unlike predictive models, which estimate a value, generative models create new artifacts, such as:
 - images
 - text
 - geometry
 - spatial layouts
- The key idea is:

The model learns a **distribution of possible designs**, and then samples from that distribution.

TYPES OF GENERATIVE MODELS

There are several major families of generative models:

- **GANs (Generative Adversarial Networks):**
Two networks compete—one generates, one evaluates.
- **VAEs (Variational Autoencoders):**
Learn a compressed latent space and generate variations.
- **Diffusion Models:**
Current state-of-the-art; generate data by gradually removing noise.
- **LLMs and Multimodal Models:**
Generate across text, image, and increasingly geometry.

TYPES OF GENERATIVE MODELS

In architecture, these models can generate:

- floor plans
- façades
- massing
- spatial configurations

DIFFUSION MODELS

Diffusion models work by:

1. Starting with random noise
 2. Iteratively removing noise
 3. Converging to a structured output
- They learn how data “looks” at different levels of noise and reconstruct it step by step.

DIFFUSION MODELS

- Their strength is:
 - high visual quality
 - stability
 - controllability through prompts or conditions

This is why most current image-based AI tools rely on diffusion.

TEXT-TO-IMAGE MODELS

- Text-to-image models allow us to generate images from textual descriptions.

For example:

“Mid-rise housing with green terraces and strong daylight access”

- The model **translates semantic intent to visual form.**

TEXT-TO-IMAGE MODELS

In architecture:

- early ideation
- atmosphere and concept exploration
- visual narratives

PROMPTING AS DESIGN

Prompting becomes a form of **design specification**.

A prompt encodes:

- intent
- constraints
- style
- performance hints

For example, adding:

- “passive cooling”
- “south-facing shading”

begins to embed **design logic into language**.

PROMPTING AS DESIGN

- iterative
- sensitive to wording
- not fully controllable

LIMITATIONS OF PROMPTING

Pure prompting has key limitations:

- limited geometric control
- no direct performance evaluation
- biases from training data
- often produces non-buildable outputs

So while it is powerful for exploration, it is **insufficient** for **architectural decision-making**.

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This is why we need integration with:

- parametric systems
- simulation tools
- structured representations

FROM IMAGES TO ARCHITECTURE

One of the biggest challenges is:

- AI generates **images**, but Architects need **geometry + systems**

Bridging strategies include:

- converting images into geometry (segmentation, tracing)
- mapping latent representations into parametric systems
- using graph-based spatial representations

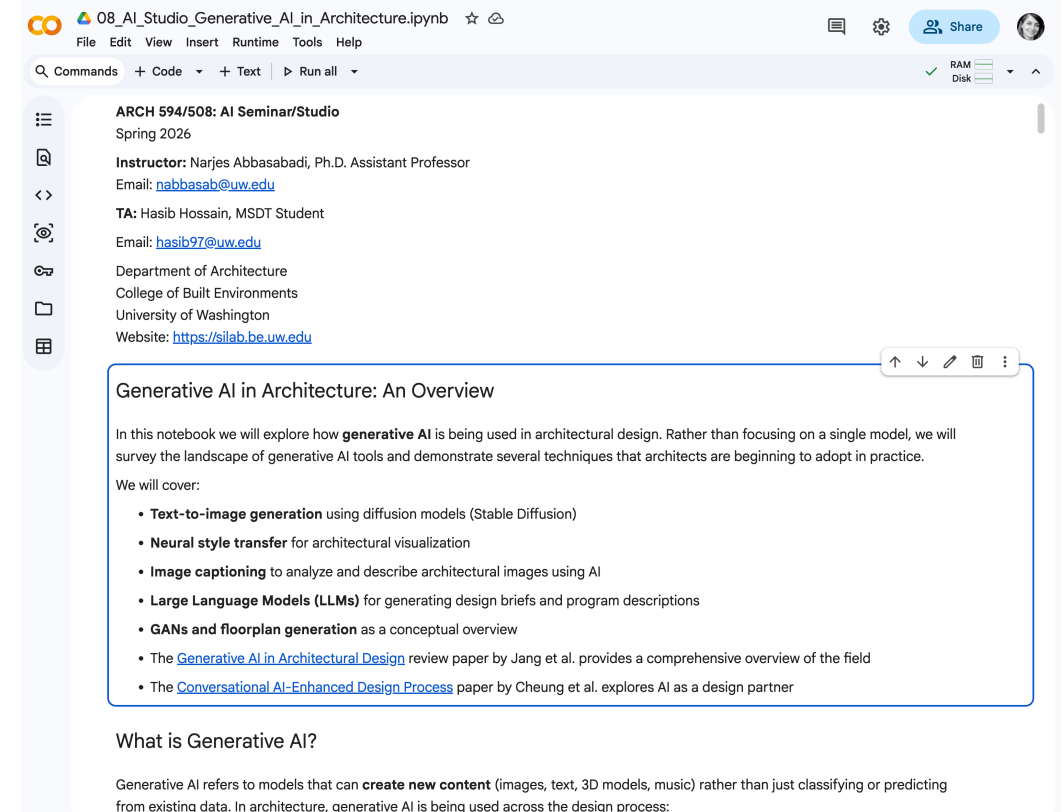
This translation step is critical.

WORKFLOW FROM THE NOTEBOOK

Your notebook demonstrates a typical generative workflow:

1. Define a prompt or condition
2. Generate outputs
3. Process outputs
4. Extract usable information
5. Integrate into a design system

Design is not a single act here....it is a **pipeline**.



CODE AS DESIGN MEDIUM

In this workflow, code becomes central.

Instead of manually crafting each design, we:

- define rules
- generate multiple outputs
- evaluate systematically

This means design becomes:

- programmable
- reproducible
- scalable

GENERATIVE AI IN ARCHITECTURAL DESIGN

- Jang, Roh, and Lee's 2025 review of 161 studies on how g pipelines, and evaluation practice.

Automation in Construction 174 (2025) 106174

161

papers reviewed

96%

gap reduction

68.9%

in schematic design

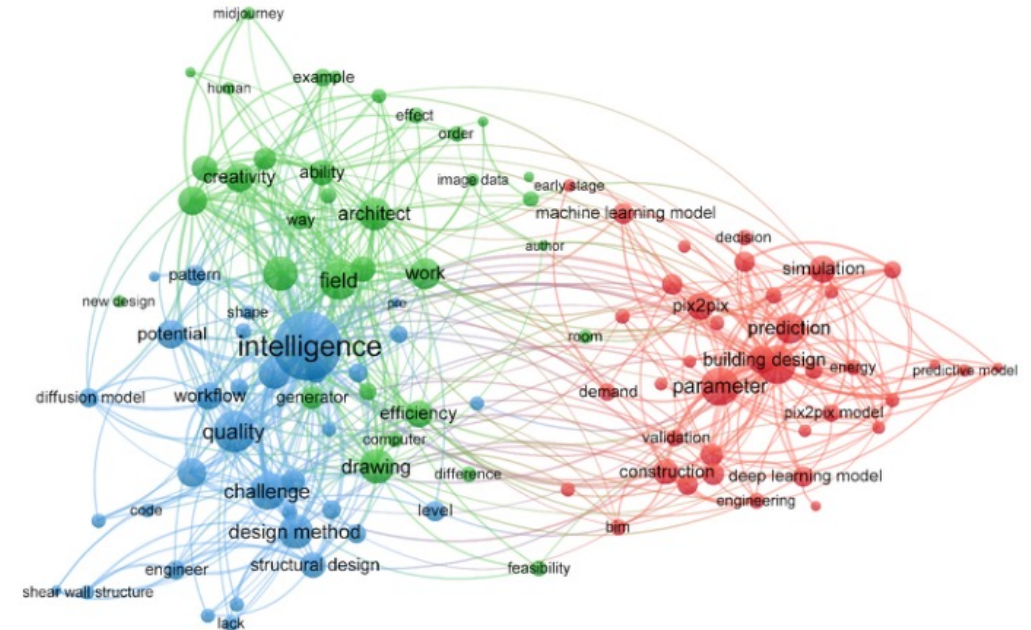


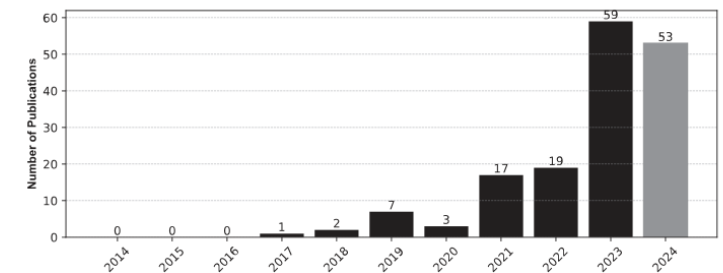
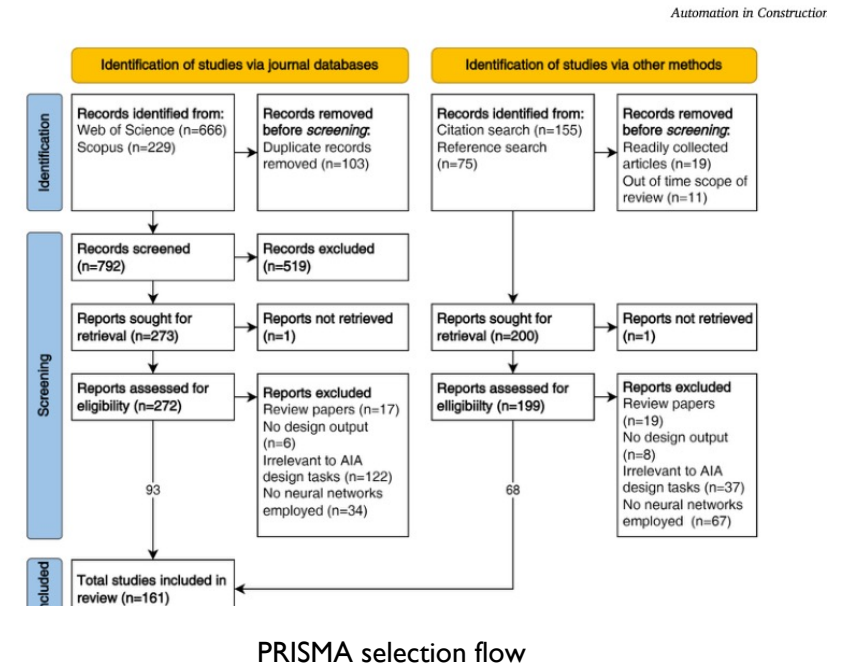
Fig. 5. Co-occurrence network of keywords from screened papers.

Keyword network from the article summarizing the field's main clusters: parameterized design, AI capability in design, and implementation challenges.

HOW THE REVIEW WAS CONDUCTED

- How fast are AI model architectures being adopted in architectural design?
- Which design phases, tasks, and objectives are receiving AI support?
- What input/output data types dominate current workflows?
- How are researchers evaluating systems when design quality is subjective?

PRISMA-guided search across WoS and Scopus, plus snowball reference/citation search. Final corpus: 161 eligible studies spanning 2014-2024.



Publication count rises sharply after 2021

AI ADOPTION IN ARCHITECTURE

- Adoption is accelerating - and no longer slowly
- The review shows a dramatic collapse in the delay between an AI model's invention and its use in architecture.
- GANs lead the literature with 81 papers (50.31%), while diffusion models (23 papers) and transformers (9 papers) rise quickly after 2022.

96.09%
reduction in theory-application
gap

62y
earliest gap

2.5y
latest gap

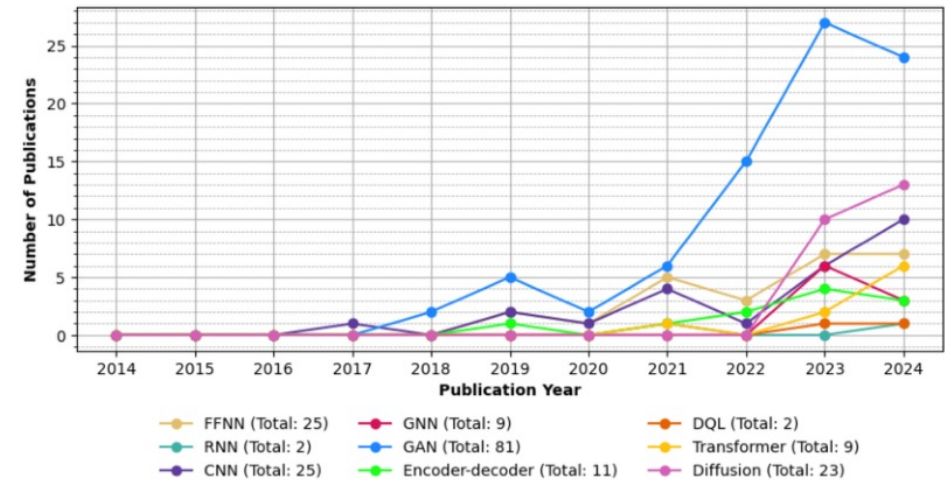
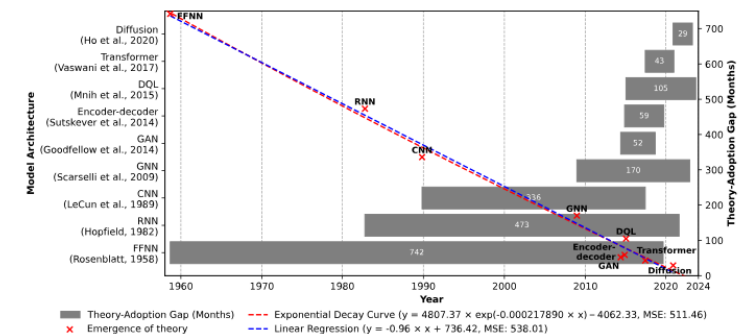


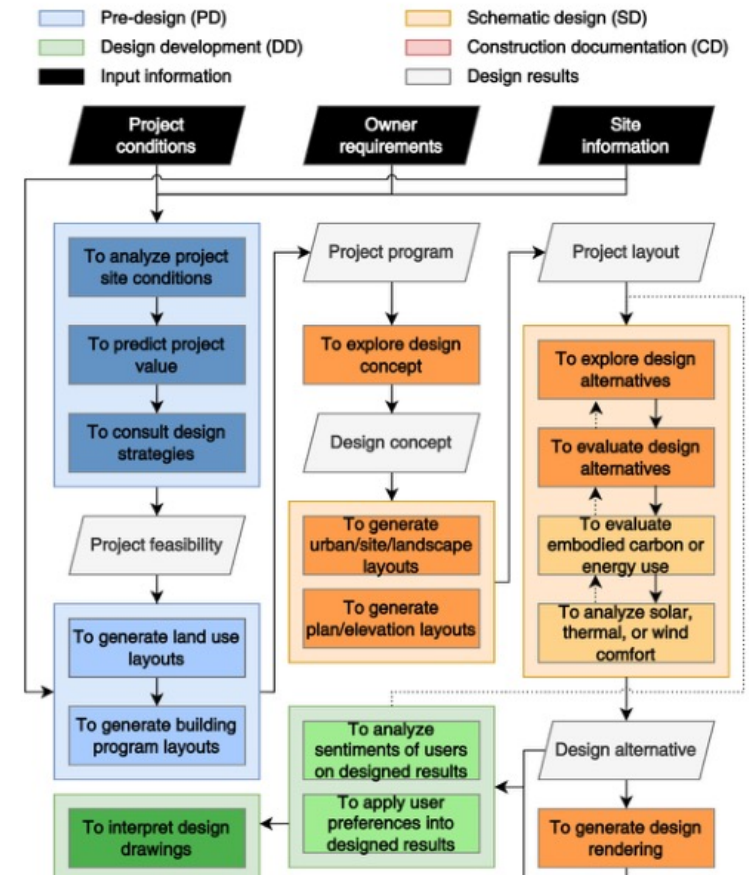
Fig. 6. Number of publications per AI model architecture.



theory-application gap of AI model architectures in architectural design (Fitted exponential decay curve: $y = 4807.37 \times \exp(-0.00021789 \times x) - 4062.33$, MSE: 511.46)
 equation: $y = -0.96 \times x + 736.42$.

WHERE AI ENTERS THE ARCHITECTURAL WORKFLOW

- AI activity is heavily front-loaded into early design.
- Most publications address conceptual exploration, optioning, rendering, and sustainability analyses in the schematic phase.
- Later phases remain thinner: DD is growing, but CD tasks such as specifications and construction drawings are still rare.



68.94%

Schematic design

21.74%

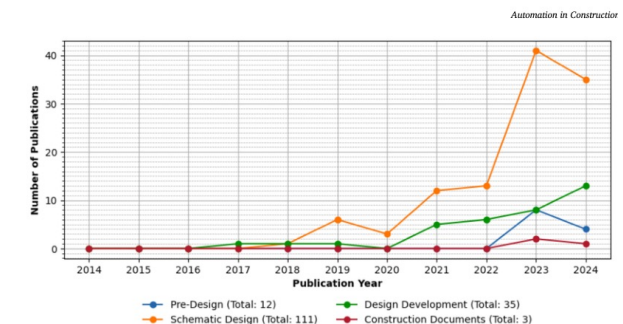
Design development

7.45%

Pre-design

1.86%

Construction docs



MOST STUDIED TASKS AND OBJECTIVES

- Generate plan/elevation layouts - 28 papers
- Explore design concepts - 21 papers
- Align architectural and structural design - 20 papers
- Analyze solar / thermal / wind comfort - 14 papers
- Evaluate embodied carbon or energy use - 13 papers

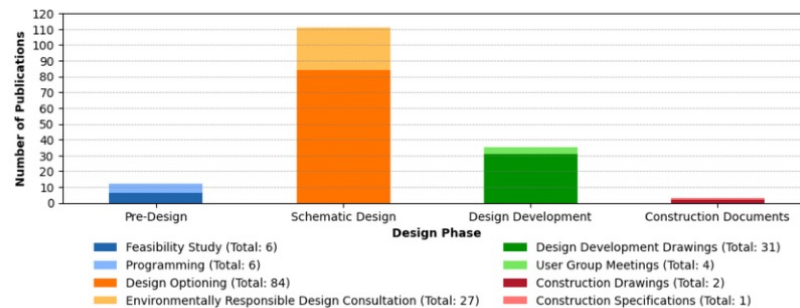
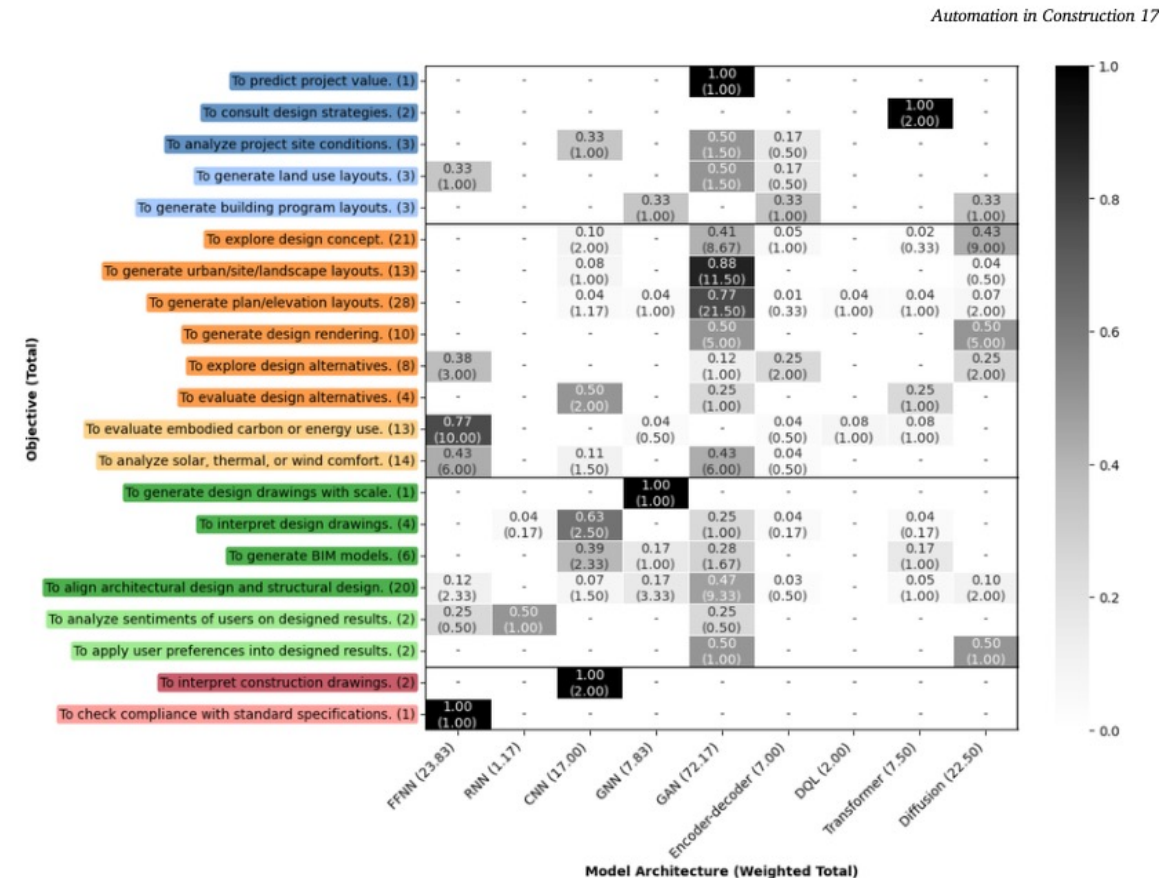


Fig. 9. Number of publications per design task in each phase.



- The most common transformation is image-to-image generation, which reflects a strong preference for planar visual outputs such as layouts, drawings, and renderings.
- Variables matter when AI is used as a surrogate model for sustainability or parameter optimization.
- Graphs and 3D models appear less often, but the authors highlight them as promising paths toward richer design semantics and more controllable workflows.

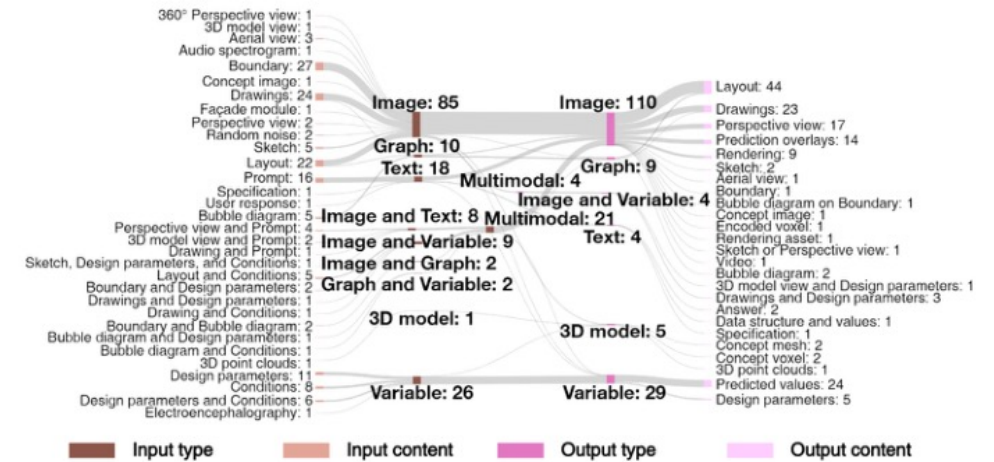
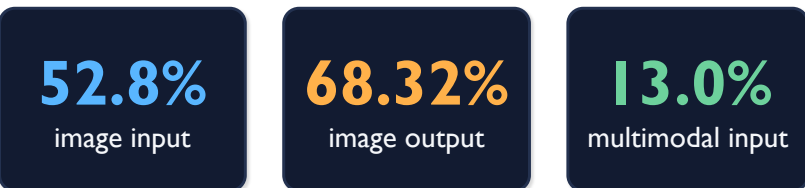
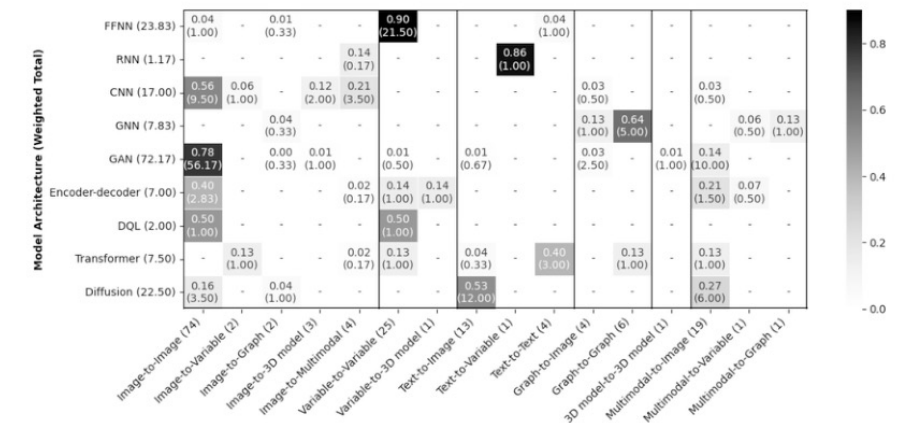


Fig. 11. Flow of input-to-output data type and content using AI in architectural design.



EVALUATION IS DIVERSE & FRAGMENTED

- Most papers rely on a single evaluation type; only 27 combine methods.
- Dataset validation is the most common method, followed by author visual interpretation.
- The review identifies 105 unique metrics across 161 studies - a warning sign that the field still lacks shared benchmarks.

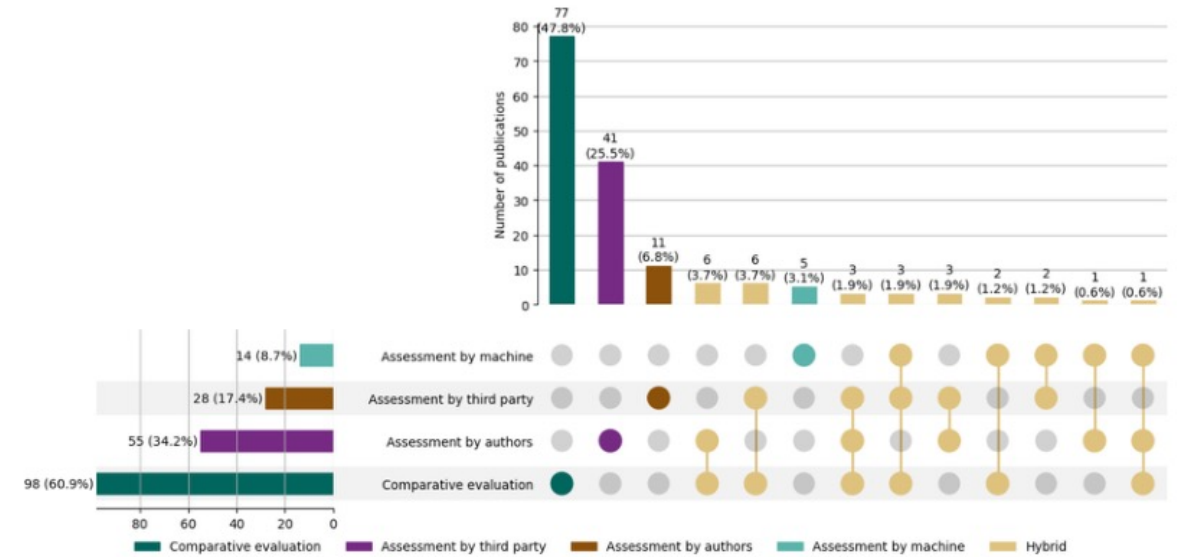
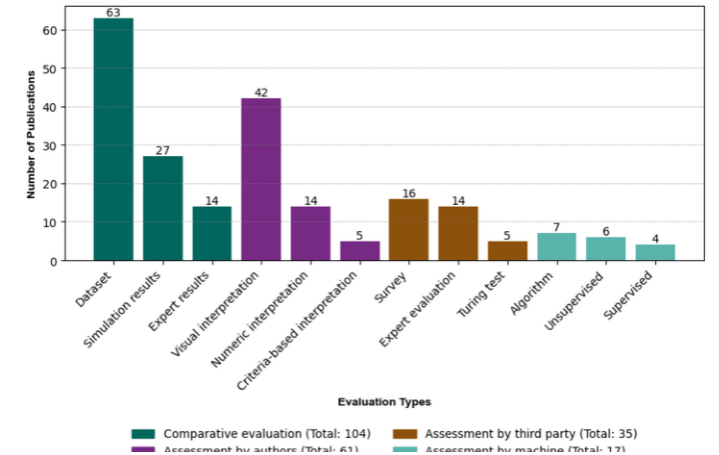


Fig. 14. Evaluation types utilized for assessing generative AI in architectural design.

60.9%
comparative

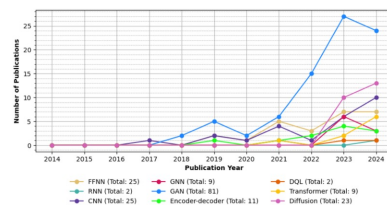
34.2%
author assessment

17.4%
third party

8.7%
machine

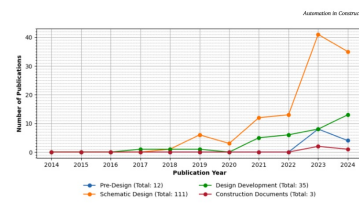
CONCLUSION

- It reduces hype by revealing where AI is genuinely mature and where later-phase claims remain thin.
- It offers a domain-specific vocabulary for discussing model architecture, design objective, data format, and evaluation strategy together.



1. Adoption timeline

Shows which model families entered architectural design and when.



2. Design process map

Locates AI activity across pre-design, SD, DD, and CD.

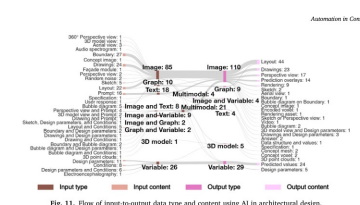


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3. Data ecology

Tracks how images, variables, text, graphs, and 3D models flow.

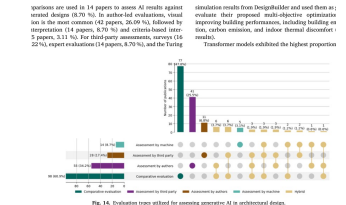


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4. Evaluation lens

Explains how researchers judge systems without a single right answer.

KEY GAPS AND FUTURE OPPORTUNITIES

- The paper ends with a clear message: AI is moving faster than the field's standards, controls, and later-phase integration.
- Generative AI in architecture is not just about novel images - it is about building reliable, controllable, benchmarked design systems.

Expand beyond early design
AI is concentrated in schematic design. DD and CD need stronger support for drawings, specs, and coordination.

Build shared benchmarks
The field needs domain datasets and metrics that measure usability, dimensional accuracy, and contextual fit - not only borrowed computer vision scores.

Improve human-in-the-loop control
Prompting alone is not enough. Designers need interfaces and models that understand intent without repeated trial and error.

Create practice-ready frameworks
Future systems should connect AI with BIM, domain knowledge, and robust workflows rather than isolated image generators.

KEY TAKEAWAYS

- Generative AI has become a serious design technology - but it is still strongest at seeing, sketching, and comparing, not yet at fully documenting buildings.
- Architectural AI adoption is accelerating fast enough that curricula and professional practice cannot treat it as a future topic.
- The current center of gravity is schematic design: concept generation, layout generation, rendering, and sustainability surrogate modeling.
- Images dominate the field, but graph, multimodal, and BIM-linked approaches point toward the next generation of more controllable systems.
- Evaluation remains the weakest common language; the field needs shared benchmarks and domain metrics.
- The real opportunity is not replacing architects - it is augmenting design reasoning with trustworthy, workflow-aware AI.

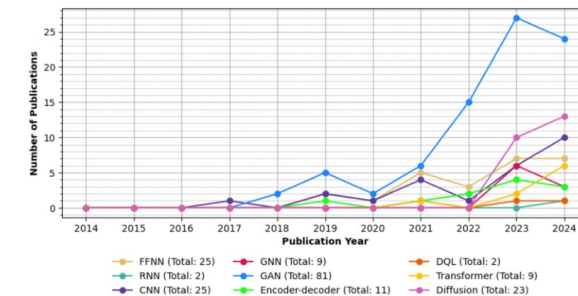
161
papers

96.09%
gap down

68.9%
SD

52.8% /
68.32%
image input / output

60.9%
comparative eval



GENERATIVE + AGENTIC AI

- The **Conversational, agentic AI-enhanced architectural design process: three approaches to multimodal AI-enhanced early-stage performative design exploration** paper by Cheung et al. explores AI as a design partner

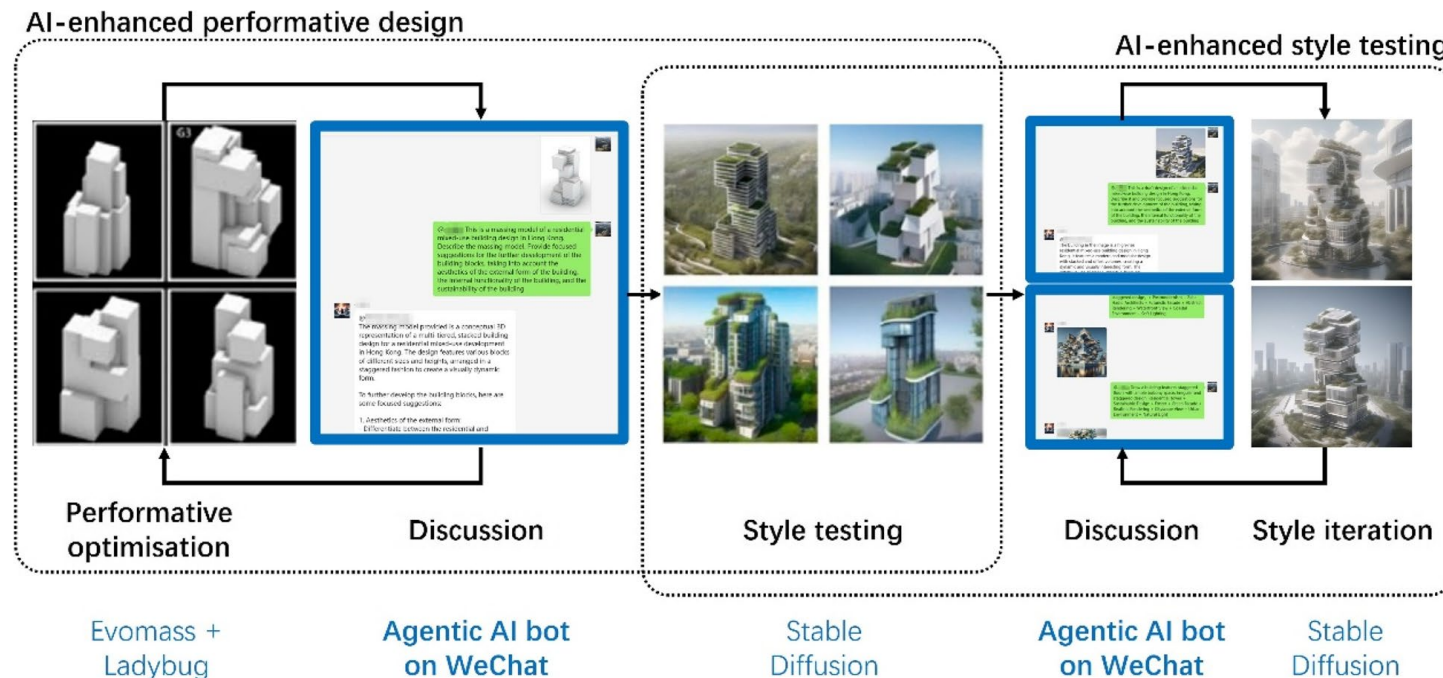


Fig. 2 Proposed 2-stage workflow: (1) AI-enhanced performative design and (2) AI-enhanced style testing

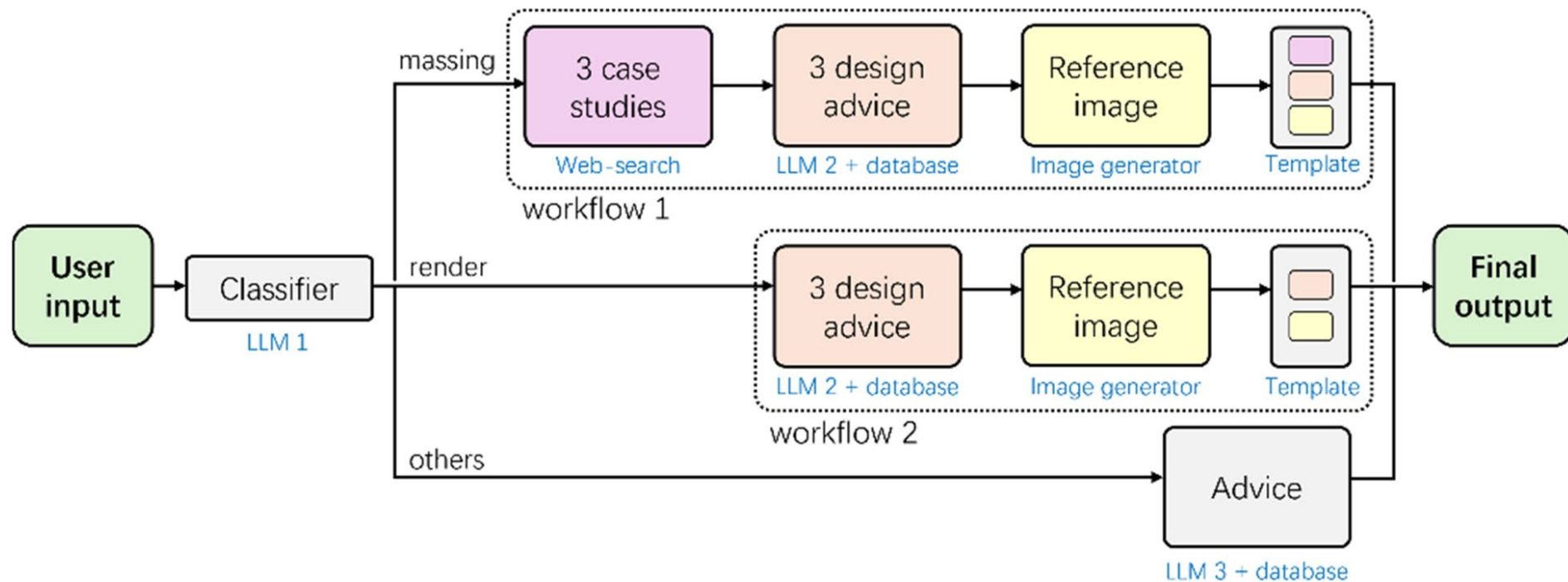


Fig. 3 The proposed AI system is composed of a classifier, two workflows and an agentic AI

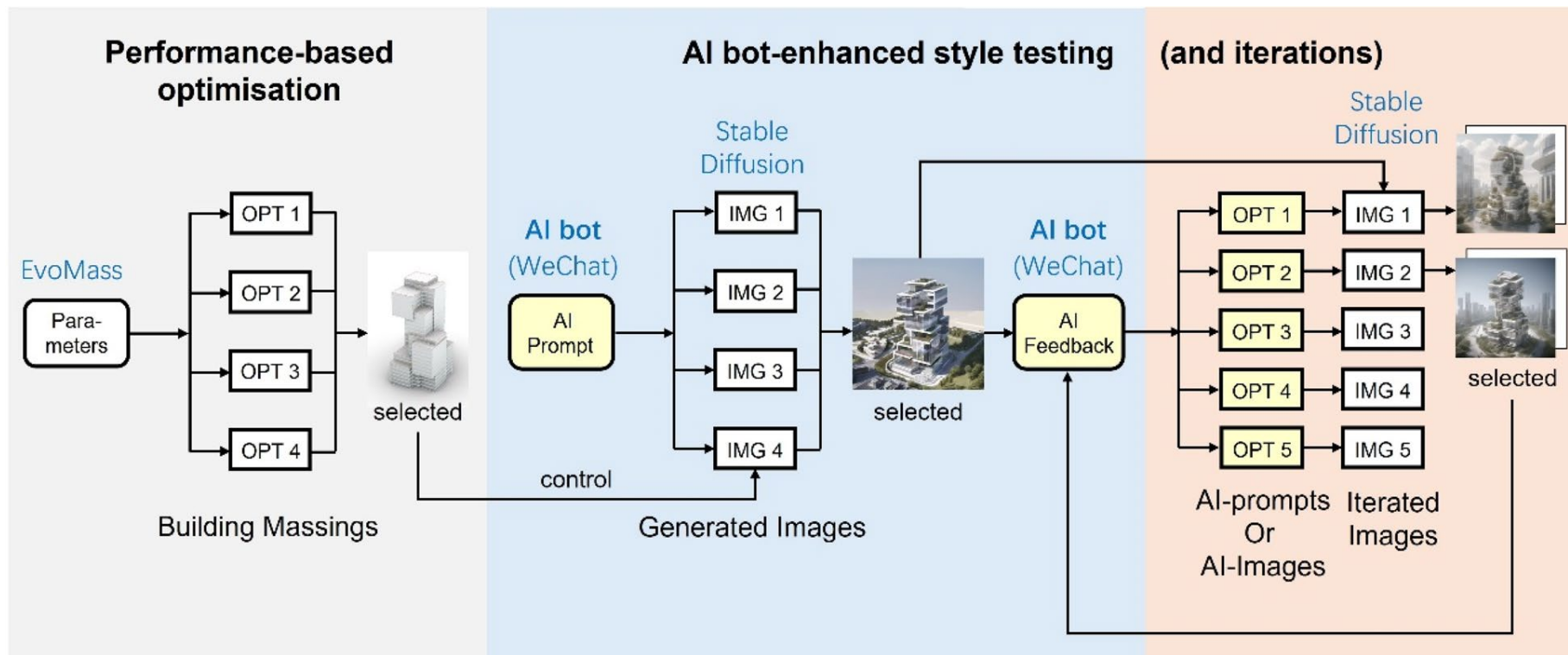


Fig. 5 A proposed conversational design workflow for the demonstration, including (1) performative optimisation, AI-enhanced style testing and (3) AI-enhanced style testing iterations

Table 1 Summary table of the AI tools, functions, design implications, and communication mediums in the demonstration

Stage	Tools	Function	Design implications	Medium/format
Performative optimisation	EvoMass and Ladybug	Massing generation with performative optimisation	Generate massing models with performative evaluations	Images (screenshot of the model)
	Developed an AI bot	Visual-question-answering	Evaluation and advice of the performative optimised massing models	Text
AI-enhanced style testing (and iterations)		Text-based conversation	Advice for style testing prompts	Text
	SD ControlNet: – MLSD	straight line detection	Suitable for style testing with highly consistent architectural form	Image
	SD ControlNet: Canny	edge detection		Image
	SD ControlNet: Depth	3D depth detection	Suitable for style testing with less design fixation, but still keeping the overall performative-optimised form	Image
	SD ControlNet: Soft Edge	line weight detection	Similar to the Depth model, but with more details and information	Image
	SD – mode: “balanced”	Balanced ControlNet guidance	Style applied, but the building form is mostly fixed	Image
	SD – mode: “prompt-focus”	Smallest ControlNet guidance	Most balanced between keeping the optimised form and design variations	Image
	SD – mode: “CN-focus”	Biggest ControlNet guidance	Most controlled of the building form, with little style testing	Image
	Developed an AI bot	Visual-question-answering	Evaluation and advice of the performative optimised massing models	Text
		Text-based conversation	Advice for style testing prompts	Text
		Image generation	Reference image generation for style testing in SD	Image

MAIN READING ASSIGNMENTS

- The **Generative AI in architectural design: Application, data, and evaluation methods** review paper by Jang et al. provides a comprehensive overview of the field:

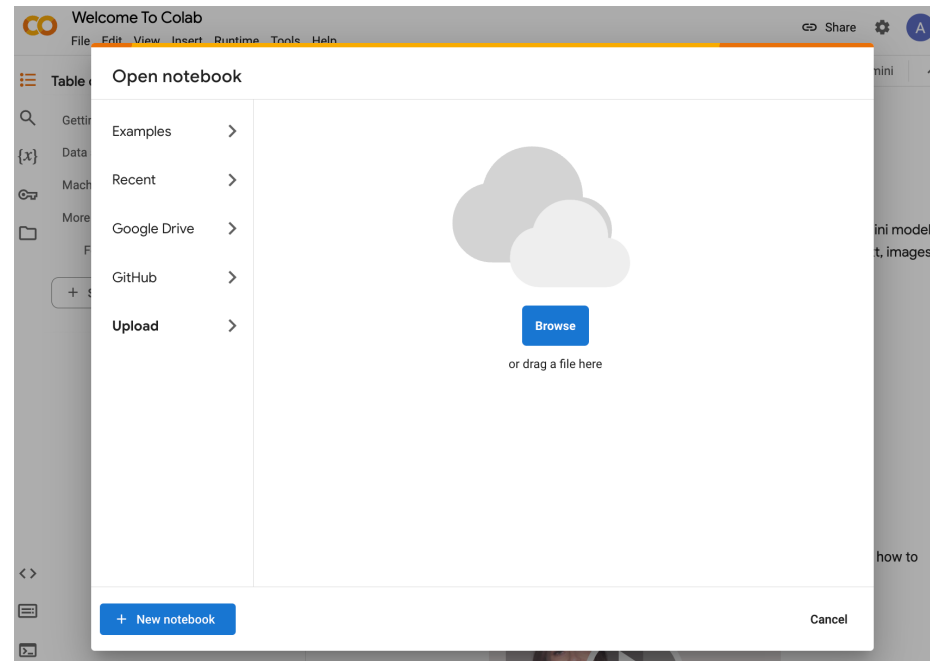
<https://www.sciencedirect.com/science/article/pii/S0926580525002146>

- The **Conversational, agentic AI-enhanced architectural design process: three approaches to multimodal AI-enhanced early-stage performative design exploration** paper by Cheung et al. explores AI as a design partner

<https://link.springer.com/article/10.1007/s44223-025-00092-5>

LAB ACTIVITY: GENERATIVE AI

1. Open up a web browser. Google “google colab” click on the first entry. You may also access the website by following <https://colab.research.google.com/>
2. Go to Canvas and download the file called “AI_Studio-Seminar_Generative_AI_in_Architecture.ipynb”



NEXT CLASS

- Agentic AI

